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False discovery rate control for high dimensional networks of quantile associations conditioning on covariates

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Summary. Motivated by gene coexpression pattern analysis, we propose a novel sample quantile contingency (SQUAC) statistic to infer quantile associations conditioning on covariates. It features enhanced flexibility in handling variables with both arbitrary distributions and complex association patterns conditioning on covariates. We first derive its asymptotic null distribution, and then develop a multiple-testing procedure based on the SQUAC statistic to test simultaneously the independence between one pair of variables conditioning on covariates for all $p(p-1)/2$ pairs. Here, p is the length of the outcomes and could exceed the sample size. The testing procedure does not require resampling or perturbation and thus is computationally efficient. We prove by theory and numerical experiments that this testing method asymptotically controls the false discovery rate. It outperforms all alternative methods when the complex association patterns exist. Applied to a gastric cancer data set, this testing method successfully inferred the gene coexpression networks of early and late stage patients. It identified more changes in the networks which are associated with cancer survivals. We extend our method to the case that both the length of the outcomes and the length of covariates exceed the sample size, and show that the asymptotic theory still holds.

Keywords: False discovery rate; Gene coexpression networks; High dimensional networks; Quantile regression

1. Introduction

In recent years, inference on high dimensional association networks has attracted considerable attention. Let $\mathbf{Y}$ represent a high dimensional multivariate outcome. The goal is to model the association between elements of $\mathbf{Y}$, without or with covariates $\mathbf{X}$. Linear associations or rank associations have been studied in many existing references. See, for example, Drton and Perlman (2007), Li *et al.* (2012), Cai, Li, Liu and Xie (2013), Chen *et al.* (2016) and Cai and Liu (2016). Beyond linear or rank associations, more general associations without parametric distribution assumptions have been studied. Chun *et al.* (2016) used joint quantile regression with penalization to estimate graphical models. Voorman *et al.* (2014) proposed an additive model to describe the relationship between variables and estimated their graphical structure by using basis function expansion and penalized regression. Li *et al.* (2014) proposed a similar

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model to study the graphical structure that is induced by additive conditional independence and focused on its theoretical features. Although these methods modelled general associations, they did not allow for adjusting for covariates $\mathbf{X}$. In addition, their general association estimators have relatively complex forms so their asymptotic null distributions are difficult to derive. As a result, it is difficult to control the type I error when inferring general associations.

An example of the general association network is the gene coexpression network. In gene coexpression network analysis, high dimensional gene expression data (microarray or ribonucleic acid sequencing) are collected, often together with other covariates. Three main challenges, among others, exist in the analysis. First, covariates (such as gender, race and cancer stage) may affect the expression distributions or distort their association patterns. Therefore, ignoring these covariates will lead to both false discoveries and false non-discoveries in network inference. Second, gene expressions cannot always be normalized into Gaussian or commonly seen parametric distributions. To model the coexpression pattern properly, we need a model that allows flexible expression distributions and association patterns. Third, a type I error measure is often desired for the analysis so that researchers can understand how reliable the inferred associations are and can prioritize the validation studies. To the best of our knowledge, no existing method can address these three challenges simultaneously.

To address these challenges, we develop a robust and computationally efficient multiple-testing procedure to infer a high dimensional sparse general association network conditioning on covariates. Our method uses conditional quantile associations to capture a wide range of general association patterns. The test statistic is called the sample quantile contingency (SQUAC) statistic. It has a neat asymptotic null distribution, enabling us to conduct computationally efficient multiple testing with accurate false discovery rate (FDR) control.

The rest of the paper is organized as follows. In Section 2, we propose a novel summary statistic to evaluate the pairwise quantile association conditioning on covariates. We also propose a multiple-testing procedure that controls the FDR. In Section 3, we prove that the test statistic converges to a χ^2 -distribution under the null, and the FDR control procedure is valid. In Section 4, we design numerical experiments to compare the multiple-testing method proposed and several alternative methods. A real data analysis is conducted in Section 5 to investigate the dependence patterns among patients with early stage and late stage gastric cancer. In Section 6, we discuss the case with high dimensional sparse covariates. Further discussions are provided in Section 7.

The programs that were used to analyse the data can be obtained from

<http://wileyonlinelibrary.com/journal/rss-datasets>

2. Method

The following notation is used in the paper. For any $a \in \mathbb{R}$, $[a]$ represents the largest integer that is smaller than or equal to a . For two numbers $a, b \in \mathbb{R}$, $a \vee b = \max(a, b)$ and $a \wedge b = \min(a, b)$. For a vector $\mathbf{a} = (a_1, \dots, a_n)' \in \mathbb{R}^n$, let $\|\mathbf{a}\|_p = (\sum_{i=1}^n a_i^p)^{1/p}$, for any $p > 0$. Also let $\|\mathbf{a}\|_0 = \sum_{i=1}^n I(a_i \neq 0)$. For a symmetric matrix A , denote by $\lambda_{\max}(A)$ its largest eigenvalue. For a set $\mathbb{A}$, let $\text{Card}(\mathbb{A})$ represent its cardinality. For two sequences of real numbers $\{a_n\}$ and $\{b_n\}$, write $a_n = O(b_n)$ if there is a constant C such that $a_n \leq Cb_n$ holds for all sufficiently large n and write $a_n = o(b_n)$ if $\lim_{n \rightarrow \infty} a_n/b_n = 0$. If $a_n = O(b_n)$ and $b_n = O(a_n)$, then $a_n \asymp b_n$. If $\lim_{n \rightarrow \infty} a_n/b_n = 1$, write $a_n \sim b_n$. For a sequence of random variables $\{X_n\}$, and a sequence of real numbers $\{a_n\}$, if there is a constant M such that for any $\varepsilon > 0$ $\lim_{n \rightarrow \infty} P(|X_n/a_n| > M) < \varepsilon$ for sufficiently large n , write $X_n = O_p(a_n)$.

2.1. Sample quantile-based contingency table statistics

Let $\mathbf{Y} = (Y_1, \dots, Y_p)'$ be the outcome vector and $\mathbf{X} = (X_1, \dots, X_{p_x})'$ be the covariate vector. For convenience, assume that $X_1 \equiv 1$ is the intercept term. We are interested in testing the null hypotheses $H_{0,ij}: Y_i \perp\!\!\!\perp Y_j | \mathbf{X}$ versus $H_{1,ij}: Y_i \not\perp\!\!\!\perp Y_j | \mathbf{X}$. In a special case where no covariates exist, let $X = 1$ and then the null hypothesis becomes $H_{0,ij}: Y_i \perp\!\!\!\perp Y_j$.

Motivated by the commonly used Pearson χ^2 independence test, we propose a new quantile association test based on quantile regressions. Let $0 = \tau_0 < \tau_1 < \dots < \tau_{D-1} < \tau_D = 1$ be an increasing sequence of probability points. The probability jump between the two quantile points is $\nu_s = \tau_s - \tau_{s-1}$, for $s \in \{1, \dots, D\}$. Let $Q_{i,s}$ be the τ_s th conditional quantile of Y_i given $\mathbf{X}$. For convenience, we set $Q_{i,0} = -\infty$ and $Q_{i,D} = \infty$. We model the relationship between Y_i and $\mathbf{X}$ by a quantile regression model,

$$Q_{i,s} = \mathbf{X}'\beta_{0,i}(\tau_s), \quad s \in \{1, \dots, D-1\}. \quad (1)$$

Quantile regression is more flexible than linear regression, because the coefficients $\beta_{0,i}(\tau_s)$ can differ across different quantile levels τ_s . Suppose that, for the k th subject, we observe $\mathbf{Y}_k = (Y_{k1}, \dots, Y_{kp})'$ and $\mathbf{X}_k = (X_{k1}, \dots, X_{kp_x})'$ as a realization of the outcome and the covariate, $k \in \{1, \dots, n\}$. When $p_x \ll n$, the quantile regression coefficients can be estimated consistently (Koenker, 2005) by

$$\hat{\beta}_i(\tau_s) = \arg \min_{\beta \in \mathbb{R}^{p_x}} \sum_{k=1}^n \{\rho_{\tau_s}(Y_{ki} - \mathbf{X}_k'\beta)\}, \quad s \in \{1, \dots, D-1\}, \quad (2)$$

with the loss function $\rho_{\tau_s}(y) = y\{\tau_s - I(y < 0)\}$. Subsequently, the conditional τ_s th sample quantile is estimated by $\hat{Q}_{k,i,s}^* = \mathbf{X}_k'\hat{\beta}_i(\tau_s)$, for $s \in \{1, \dots, D-1\}$, $\hat{Q}_{k,i,0}^* = -\infty$ and $\hat{Q}_{k,i,D}^* = \infty$.

In reality, the randomness in quantile fitting might result in quantile level crossing problems: there might be some $\hat{Q}_{k,i,s}^* > \hat{Q}_{k,i,s+1}^*$, for some $k \in \{1, \dots, n\}$, $i \in \{1, \dots, p\}$ and $s \in \{1, \dots, D-2\}$. Many references have discussed how to modify the quantile regression fitting to obtain non-crossing estimated quantiles. See, for example, He (1997), Neocleous and Portnoy (2008) and Bondell *et al.* (2010). In this paper, to minimize the conditions that are needed and to simplify the computation procedure, we propose a swapping approach to solve the possible quantile level crossing problem.

After deriving the estimated conditional quantiles $\hat{Q}_{k,i,s}^* = \mathbf{X}_k'\hat{\beta}_i(\tau_s)$, for $s \in \{1, \dots, D-1\}$, we start from the beginning of the sequence and check the adjacent estimators. For $s = 1, \dots, D-2$, if $\hat{Q}_{k,i,s}^* > \hat{Q}_{k,i,s+1}^*$, we swap them so that $\hat{Q}_{k,i,s}^* = \hat{Q}_{k,i,s+1}^*$ and $\hat{Q}_{k,i,s+1}^* = \hat{Q}_{k,i,s}^*$; otherwise let $\hat{Q}_{k,i,s}^* = \hat{Q}_{k,i,s}^*$ and $\hat{Q}_{k,i,s+1}^* = \hat{Q}_{k,i,s+1}^*$. In this way, the estimated quantile levels will not cross. In Section 3, we shall show that the swapping approach will maintain the theoretical properties that are needed for proving the theoretical results in the paper.

For each pair of the outcome variables (Y_i, Y_j) , we then construct a $D \times D$ SQUAC table. The (s, t) th cell of the table is the counts

$$O_{ij,st} = \sum_{k=1}^n I(\hat{Q}_{k,i,s-1}^* < Y_{ki} \leq \hat{Q}_{k,i,s}^*) I(\hat{Q}_{k,j,t-1}^* < Y_{kj} \leq \hat{Q}_{k,j,t}^*).$$

It is easy to see that if all the conditional quantiles $Q_{k,i,s}$ are known, under $H_{0,ij}$, the expected number of observations that will fall in the (s, t) th cell is $E_{st} = n\nu_s\nu_t$. Inspired by the Pearson χ^2 -test, we propose the SQUAC test statistic

$$T_{ij} = \sum_{s=1}^D \sum_{t=1}^D \frac{(O_{ij,st} - E_{st})^2}{E_{st}}. \quad (3)$$

Although T_{ij} and the Pearson χ^2 -test statistic look similar in format, they are fundamentally different. The traditional Pearson χ^2 -test statistic is designed for testing independence between categorical variables, so that the cell boundaries of the contingency table are pre-fixed. For a SQUAC table, because the cell boundaries depend on quantile regression estimators, they are different across samples and, for each sample, the cell boundaries vary when others samples change. When deriving the asymptotic properties of T_{ij} and the testing procedure, we shall include the variation that is introduced by the conditional quantile point estimate.

2.2. False discovery rate control procedure

By bounding the errors between the estimated conditional quantiles and the true quantiles and projection techniques, we shall show that, under $H_{0,ij}$, T_{ij} asymptotically follows $\chi^2\{(D-1)^2\}$. Under the alternative hypothesis $H_{1,ij}$, for any constant C , as n and p increase, the probability that $T_{ij} \leq C$ will be vanishing (see theorem 1 and corollary 2, and their proofs for details). On the basis of these results, we construct an FDR control procedure to test $H_{0,ij}$ ($1 \leq i < j \leq p$) simultaneously. We reject $H_{0,ij}$ if $T_{ij} > t$ and determine the threshold t on the basis of the desired FDR level α .

Define the null set $\mathbb{H}_0 = \{(i, j) : 1 \leq i < j \leq p, H_{0,ij} \text{ is true}\}$, and the full set $\mathbb{H} = \{(i, j) : 1 \leq i < j \leq p\}$. The alternative set is $\mathbb{H}_1 = \mathbb{H} \setminus \mathbb{H}_0$. Let

$$\begin{aligned} q &= \text{Card}(\mathbb{H}) = p(p-1)/2, \\ q_0 &= \text{Card}(\mathbb{H}_0). \end{aligned} \quad (4)$$

The false discovery proportion (FDP) and FDR are defined as

$$\begin{aligned} \text{FDP} &= \frac{\sum_{(i,j) \in \mathbb{H}_0} I(T_{ij} > t)}{\max \left\{ \sum_{(i,j) \in \mathbb{H}} I(T_{ij} > t), 1 \right\}} = \frac{\sum_{(i,j) \in \mathbb{H}_0} I(T_{ij} > t)/q_0}{\max \left\{ \sum_{(i,j) \in \mathbb{H}} I(T_{ij} > t), 1 \right\}} \frac{q_0}{q}, \\ \text{FDR} &= E(\text{FDP}). \end{aligned} \quad (5)$$

Since $\mathbb{H}_0$ is not known, we replace $\sum_{(i,j) \in \mathbb{H}_0} I(T_{ij} > t)/q_0$ with $G_D(t)$, the complementary cumulative distribution function (CDF) of $\chi^2\{(D-1)^2\}$. We further assume that $q_0/q \rightarrow 1$ as $p \rightarrow \infty$. It then leads to the following testing procedure.

Let $t_p = 4 \log(n \vee p) + \{(D-1)^2 - 2\} \log\{\log(n \vee p)\}$, and

$$\hat{t} = \inf \left[0 \leq t \leq t_p : \frac{q G_D(t)}{\max \left\{ \sum_{1 \leq i < j \leq p} I(T_{ij} > t), 1 \right\}} \leq \alpha \right]. \quad (6)$$

If $\hat{t}$ does not exist, let $\hat{t} = t_p$. For $1 \leq i < j \leq p$, reject $H_{0,ij}$ if $T_{ij} > \hat{t}$.

The search range of $\hat{t}$ is restricted to $[0, t_p]$ because, by the proof of theorem 3, $\lim_{n,p \rightarrow \infty} P(\sup_{(i,j) \in \mathbb{H}_0} T_{ij} > t_p) = 0$. In practice, to find $\hat{t}$, we need to search only at the realization values of that T_{ij} in the interval $[0, t_p]$.

We provided an equivalent form in algorithm 1 (Table 1). In algorithm 1, $\widehat{\text{FDR}}_l$ can be viewed as a scaled p -value. If $\widehat{\text{FDR}}_l \leq \alpha$, the surrogate p -value $G_D(T_{h(l)}) \leq (r+l)\alpha/q$. Therefore, algorithm 1 is essentially a Benjamini–Hochberg-type procedure (Benjamini and Hochberg, 1995) with prethresholding. The computational complexity is $O\{Dnp + p^2 \log(n \vee p)\}$. Considering that the total number of pairs (Y_i, Y_j) is $p(p-1)/2 = O(p^2)$, this algorithm is very efficient. To

Table 1. Algorithm 1: FDR control algorithm

Input: SQUAC statistics T_{ij} , $(i, j) \in \mathbb{H}$
 Set the rejection set $\mathcal{R} = \emptyset$, the candidate set $\mathcal{V} = \emptyset$ and the existing label state to 0
 For $(i, j) \in \mathbb{H}$,
 if $T_{ij} > t_p$, $\mathcal{R} = \mathcal{R} \cup \{(i, j)\}$;
 otherwise $\mathcal{V} = \mathcal{V} \cup \{(i, j)\}$
 Set $r = \text{Card}(\mathcal{R})$
 Rank $T_{h(1)} \geq T_{h(2)} \geq \dots \geq T_{h(q-r)}$, where $h: \{1, \dots, q-r\} \rightarrow \mathcal{V}$ is the corresponding index mapping
 Set $l^* = 0$.
 While $l \in \{q-r, \dots, 1\}$ and state = 0,
 set $\text{FDR}_l = qG_D(T_{h(l)})/(r+l)$;
 if $\text{FDR}_l \leq \alpha$, set $l^* = l$, and state = 1,
 Set $\mathcal{R} = \mathcal{R} \cup \{h(1), \dots, h(l^*)\}$
Output: the rejection set $\mathcal{R}$

facilitate the general audience to use this method, we have provided the R code, the corresponding help file and some examples at <https://github.com/jichunxie/squac>.

3. Asymptotic properties

For any $1 \leq i \leq p$, define $s_0(i) = \text{Card}\{j: 1 \leq j \leq p, j \neq i, Y_i \not\perp Y_j | \mathbf{X}\}$, and $d_p = \max_{1 \leq i \leq p} s_0(i)$. If we treat each variable Y_i as a node and the dependence between a pair of variables as an edge, in the resulting network, d_p is the maximum degree. To capture the level of association in the (s, t) th cell of the SQUAC table, let

$$\gamma_{ij,st} = E\{[I(Q_{k,i,s-1} < Y_{ki} \leq Q_{k,i,s}) - \nu_s]\{I(Q_{k,j,t-1} < Y_{kj} \leq Q_{k,j,t}) - \nu_t\}/(\nu_s \nu_t)^{1/2}\}.$$

It is easy to see that $\gamma_{ij,st} = 0$ if Y_i and Y_j are independent given $\mathbf{X}$. For any constant $M > 0$, define the high quantile association set as

$$\mathbb{B}_M = \{(i, j): 1 \leq i, j \leq p, \exists (s, t), \text{ such that } |\gamma_{ij,st}| \geq M\{\log(n \vee p)/n\}^{1/2}\}. \quad (7)$$

To derive the asymptotic properties, we need the following conditions.

Condition 1. Let $f_{\mathbf{X},i}$ be the conditional probability density function of $Y_i | \mathbf{X}$. Assume that $|f_{\mathbf{X},i}(y)| \leq C_1$, $\forall y \in \mathbb{R}$, $\mathbf{X} \in \mathbb{R}^{p_x}$, and $i = 1, \dots, p$.

Condition 2. Assume that p_x is a constant. And also there are a constant C_2 and some small positive constant ε such that

$$P\left\{\sup_{s=1}^D |\hat{Q}_{k,i,s} - Q_{k,i,s}| > 2C_2(1+\varepsilon)n^{-1/2} \log(n \vee p)^{1/2}\right\} < D(n \vee p)^{-2-\varepsilon}. \quad (8)$$

Condition 3. Suppose that $p \leq cn^r$, for some $r > 0$. Also suppose that there is some $C_3 > 0$ such that $D \leq C_3[\log(n \vee p)/\log\{\log(n \vee p)\}]^{1/2}$.

Condition 4. Let $0 < u_0 = \min_{1 \leq s \leq D} \nu_s \leq \max_{1 \leq s \leq D} \nu_s = u_1 < 1$. Define constants $M_1 = (1 - u_0)(1 - u_0 + u_1)/u_0$, $M_2 = (3 - 2u_0)(4C_1C_2 + 2)/u_0$ and $M_3 = 4.02 + 2M_1 + M_2$. Consider $\mathbb{B}_M$ defined in expression (7) with $M = M_3$. Let $c_p = \text{Card}(\mathbb{B}_{M_3})$. We assume that $d_p = o(\sqrt{c_p})$.

Condition 1 requires the probability density function of $Y_i|\mathbf{X}$ to be bounded. It is a mild condition satisfied by a large family of distributions.

Condition 2 requires that the estimated quantile levels achieve a certain rate of convergence. This condition is also easily satisfied. In traditional quantile regression, under mild conditions, the Bahadur representation $n^{1/2}\{\hat{\beta}_i(\tau) - \beta_{0,i}(\tau)\} = \mathbf{D}_i(\mathbf{X}, \tau)\mathbf{W}_i(\tau) + \mathbf{R}_i$ holds, where $\mathbf{D}_i(\mathbf{X}, \tau) = E[\mathbf{X}\mathbf{X}' f_{\mathbf{X},i}\{\mathbf{X}'\beta_{0,i}(\tau)\}]^{-1}$, $\mathbf{W}_i(\tau)$ is a Brownian bridge and $\mathbf{R}_i$ is an error term. It satisfies that, for any constant $K > 0$,

$$P[\|\mathbf{R}_i\|_2 = O\{n^{-1/2} \log(n)^{3/2}\}] \geq n^{-K}. \quad (9)$$

See result 1 in the appendix of Portnoy (2012).

If $\sup_{k=1}^n \|\mathbf{X}_k\|_2 \leq C$, $\sup_{i=1}^p \sup_{s=1}^D \lambda_{\max}\{\mathbf{D}_i(\mathbf{X}, \tau_s)\} \leq C$, and $p \leq cn^r$ for some $r > 0$, then for some sufficiently large C_2 we have

$$\begin{aligned} P\{\|\hat{Q}_{k,i,s}^* - Q_{k,i,s}\| > 2C_2(1+\varepsilon)n^{-1/2} \log(n \vee p)^{1/2}\} \\ &= P\{n^{1/2}|\mathbf{X}'_k \hat{\beta}_i(\tau_s) - \mathbf{X}'_k \beta_{i,0}(\tau_s)| > 2C_2(1+\varepsilon) \log(n \vee p)^{1/2}\} \\ &\leq P[\|\mathbf{X}_k\|_2 \|\mathbf{D}_i(\mathbf{X}, \tau_s)\mathbf{W}_i(\tau_s) + \mathbf{R}_i\|_2 > 2C_2\{(1+\varepsilon) \log(n \vee p)\}^{1/2}] \\ &\leq P\left[\frac{C\|\mathbf{D}_i(\mathbf{X}, \tau_s)\mathbf{W}_i(\tau_s)\|_2}{C_2} > 2\{(1+\varepsilon) \log(n \vee p)\}^{1/2} - 1\right] + P\left(\|\mathbf{R}_i\|_2 > \frac{C_2}{C}\right) \end{aligned} \quad (10)$$

$$< 2(n \vee p)^{-2-2\varepsilon} + (n \vee p)^{-2-2\varepsilon} < (n \vee p)^{-2-\varepsilon}. \quad (11)$$

Term (10) leads to term (11) because, when C_2 is sufficiently large, $\|\mathbf{D}_i(\mathbf{X}, \tau_s)\mathbf{W}_i(\tau_s)\|_2/C_2$ follows a Gaussian distribution with mean 0 and variance less than or equal to 1. Then by Gaussian tail probability, we have the large deviation probability controlled for the first part. For the second part, we use condition (9) and take $K = (2+2\varepsilon)(r \vee 1)$. It then follows that

$$P\left\{\sup_{s=1}^D |\hat{Q}_{k,i,s}^* - Q_{k,i,s}| > 2C_2(1+\varepsilon)n^{-1/2} \log(n \vee p)^{1/2}\right\} < D(n \vee p)^{-2-\varepsilon}, \quad (12)$$

where we define $\hat{Q}_{k,i,D}^* = Q_{k,i,D} = \infty$ and $\hat{Q}_{k,i,D}^* - Q_{k,i,D} = 0$. Although we considered the convergence of $\hat{Q}_{k,i,s}^*$ under the conditions

$$\sup_{k=1}^n \|\mathbf{X}_k\|_2 \leq C$$

and

$$\sup_{i=1}^p \sup_{s=1}^D \lambda_{\max}\{\mathbf{D}_i(\mathbf{X}, \tau_s)\} \leq C,$$

these conditions are sufficient instead of necessary. It is possible that, under weaker conditions, result (12) still holds. For example, the condition $\sup_{k=1}^n \|\mathbf{X}_k\|_2 \leq C$ can be weakened as $P(\sup_{k=1}^n \|\mathbf{X}_k\|_2 \leq C) > 1 - n^{-3r}$. In Section 2, we proposed a swapping approach to guarantee that the estimated quantiles will be non-crossing. After swapping, $|\hat{Q}_{k,i,s} - Q_{k,i,s}| \leq \sup_{t \in \{1, \dots, D-1\}} |\hat{Q}_{k,i,t}^* - Q_{k,i,t}|$. Thus, result (12) also holds for $\hat{Q}_{k,i,s}$.

Condition 2 also ensures that p_x is finite as $n \rightarrow \infty$. We shall discuss the case $p_x \geq n$ in Section 6.

Condition 3 allows p to grow faster than n , so that the asymptotic properties hold for high dimensional data. It also indicates that, when n is sufficiently large, D can also be relatively

large so that the SQUAC statistics can capture subtle conditional quantile associations in a local region of (Y_i, Y_j) .

Condition 4 requires the preset quantile levels to be bounded away from 0 and 1, because, when quantile levels approach 0 or 1, the rate of convergence of the estimated conditional quantiles becomes slower. Although it can be relaxed to allow the quantile levels to approach 0 or 1, we keep it this way to make the presentation of the asymptotic theory simple. Condition 4 also requires a subset of dependent variables to have high quantile associations. It does not impose any lower bound condition on the minimum non-zero quantile association. The number 4.02 in the constant M_3 can be replaced by any constant that is greater than 4. The condition $d_p = o(\sqrt{c_p}) \leq o(p)$ leads to $q_0/q \rightarrow 1$ when $p \rightarrow \infty$. Here q_0 and q are defined in expression (4). The term q_0/q represents the sparsity level of the conditional quantile association network of $(Y_1, \dots, Y_p)|\mathbf{X}$. When $c_p = O(p^\alpha)$ for some $\alpha \in (1, 2)$, d_p could be much larger than $p^{1/2}$.

Theorem 1. For any positive integer D , let $G_D(t)$ be the complementary CDF of $\chi^2\{(D-1)^2\}$. Under conditions 1–3, for any constant $C_0 > 0$,

$$\sup_{0 \leq t \leq C_0 \log(n \vee p)} \sup_{(i, j) \in \mathbb{H}_0} \frac{P(T_{ij} \geq t)}{G_D(t)} \rightarrow 1. \quad (13)$$

Corollary 1. Suppose that conditions 1 and 2 hold. For any positive integer D , under the null $H_{0,ij}: Y_i \perp\!\!\!\perp Y_j | \mathbf{X}$, the SQUAC statistic T_{ij} asymptotically follows the χ^2 -distribution with degrees of freedom $(D-1)^2$.

Theorem 1 considers the fact that the conditional quantile points are estimated rather than known. It shows that the true complementary CDF function of T_{ij} converges to the χ^2 complementary CDF over the range $[0, C_0 \log(n \vee p)]$ for all $(i, j) \in \mathbb{H}_0$. The convergence at the tail is a stronger result than the general convergence in distribution. At the tail point $C_0 \log(n \vee p)$, the complementary CDF of the χ^2 -distribution is very small. The rate of convergence must be faster than the decaying rate of the complementary CDF itself to make it happen.

To prove theorem 1, we decompose $T_{ij,st} = \tilde{T}_{ij,st} + R_{ij,st}$, where

$$\tilde{T}_{ij} = \sum_{s=1}^D \sum_{t=1}^D L_{ij,st}^2, \quad L_{ij,st} = n^{-1/2} \sum_{k=1}^n \frac{(I_{k,i,s} - \nu_s)(I_{k,j,t} - \nu_t)}{(\nu_s \nu_t)^{1/2}}.$$

By the multivariate central limit theorem, we can see that $(L_{ij,11}, \dots, L_{ij,DD})$ weakly converges to a degenerate multivariate Gaussian distribution. The key is to derive the range of $[0, A_n]$, where the convergence is sufficiently fast compared with the decay of the multivariate Gaussian tail probability. For this, we use an Edgeworth expansion to show that A_n could be as large as $o(n^{1/3} D^{-4})$. Apparently, for any $C_0 > 0$, $C_0 \log(n \vee p)$ is within the range $[0, A_n]$. Then we integrate over an elliptic region to show that $\tilde{T}_{ij}$ asymptotically follows $\chi^2\{(D-1)^2\}$ under $H_{0,ij}$ and the rate of convergence is also fast over the range $[0, A_n]$. In contrast, when $t \in [0, C_0 \log(n \vee p)]$, for any constant C_0 , the extra term $R_{ij,st}$ can also be bounded by the decay rate $\log(n \vee p)^{-1}$ with high probability converging to 1. These steps build up the proof of theorem 1, which is provided in Appendix A.

Theorem 2. Suppose that conditions 1–3 hold. For any $c_1 > 0$ and $c_2 > 2M_1 + M_2 + c_1$,

$$\lim_{n \rightarrow \infty} P \left\{ \inf_{(i, j) \in \mathbb{B}_{c_2}} T_{ij} > c_1 \log(n \vee p) \right\} = 1.$$

Corollary 2. Suppose that conditions 1–3 hold. If for $(i, j) \in \mathbb{H}_1$, $|\gamma_{ij}| > (2M_1 + M_2 + c_1) \times \{\log(n \vee p)/n\}^{1/2}$,

$$\lim_{n \rightarrow \infty} P\{T_{ij} > c_1 \log(n \vee p) | (i, j) \in \mathbb{H}_0\} = 0,$$

$$\lim_{n \rightarrow \infty} P\{T_{ij} > c_1 \log(n \vee p) | (i, j) \in \mathbb{H}_1\} = 1.$$

Although corollary 2 shows that T_{ij} 's null and the alternative distributions are well separated when the alternative quantile association is high, this result is not sufficient for controlling the FDR. In practice, both large and small non-zero quantile associations may exist, and the assumption that the minimal non-zero association is greater than a certain level is often unrealistic. Theorem 3 does not rely on the condition to bound the minimum non-zero association.

Theorem 3. Under conditions 1–4, the FDR and FDP of the multiple-testing procedure (6) satisfy $\lim_{n, p \rightarrow \infty} \text{FDR} = \alpha$, and FDP/α converges to 1 in probability as $n \rightarrow \infty$.

A key step to prove theorem 3 is to show that the dependence between the T_{ij} will not affect the validity of the multiple-testing procedure. The result relies on some mild condition on the network sparsity (condition 4). Such a condition and the proof technique have been used in other references (e.g. Cai, Liu and Xia (2013)).

4. Numerical experiments

We perform extensive numerical experiments to evaluate the performance of the multiple-testing procedure proposed. Consider the model

$$Y_i = \beta_{0,i} + \beta_{1,i}X_1 + \beta_{2,i}(U_{0,i})X_2 + \sigma_i Y_{0,i}, \quad Y_{0,i} = F_Y^{-1}(U_{0,i}), \quad i \in \{1, \dots, p\}. \quad (14)$$

We simulate X_1 from the truncated Gaussian $\text{TN}(0, 0.2)$ distribution with range $[-2, 2]$, and X_2 from the binary distribution $\text{Bin}(1, 0.3)$. The coefficients $\beta_{0,i}$ and $\beta_{1,i}$ are constants, and $\beta_{2,i}(U_{0,i}) = \beta_{20,i} + U_{0,i}$. We generate $\beta_{0,i}$ from $\text{Unif}(0, 0.5)$, $\beta_{1,i}$ from $0.5N(0.1, 0.3) + 0.5N(-0.1, 0.3)$ and $\beta_{20,i}$ from $0.5N(0.1, 0.3) + 0.5N(-0.1, 0.3)$. The standard error σ_i is a constant, whose value in the simulation is generated from $\text{Unif}(0.2, 0.5)$. Based on expression (14), the τ th marginal quantile of Y_i conditioning on X_1 and X_2 is

$$Q_\tau\{Y_i | \mathbf{X} = (X_1, X_2)'\} = \beta_{0,i} + \beta_{1,i}X_1 + (\beta_{20,i} + \tau)X_2 + \sigma_i Q_\tau(Y_{0,i}).$$

Here $Q_\tau(Y_{0,i})$ is the τ th marginal quantile of $Y_{0,i}$, whose CDF is denoted by F_Y . We let F_Y take the standard Gaussian distribution under the following settings 1–5, and the standard Cauchy distribution under setting 6. A brief summary of the simulation settings is provided below, and detailed descriptions are deferred to section S3 in the on-line supplementary materials.

- (a) *Setting 1* (linear dependence; Gaussian tail): the dependence between the $Y_{0,i}$ can be fully captured by its covariance matrix.
- (b) *Setting 2* (linear dependence with outliers; Gaussian tail): this setting is the same as setting 1, except that the samples are contaminated by 10% outliers generated from the Cauchy distribution.
- (c) *Setting 3* (quadratic dependence; Gaussian tail): if $Y_{0,i}$ and $Y_{0,j}$ are dependent, their relationship follows a quadratic pattern.
- (d) *Setting 4* (dependence affected by latent variables; Gaussian tail): if $Y_{0,i}$ and $Y_{0,j}$ are dependent, their level of association is affected by some unobserved latent variables.
- (e) *Setting 5* (dependence affected by marginal values of variables; Gaussian tail): if $Y_{0,i}$ and $Y_{0,j}$ are dependent, their level of association is affected by their marginal values.
- (f) *Setting 6* (quadratic dependence; heavy tail (Cauchy)): if $Y_{0,i}$ and $Y_{0,j}$ are dependent, their relationship follows a quadratic pattern.

4.1. Numerical experiments on testing conditional quantile associations

In our first set of numerical experiments, we compare the proposed SQUAC method with the methods testing linear association based on data generated by model (14). For the SQUAC method, we consider different quantile levels $D = 3$, $D = 4$ and $D = 5$, denoted by SQUAC(3), SQUAC(4) and SQUAC(5) respectively in Table 2. All quantile points τ_i are chosen evenly based on D . We consider various settings of D to illustrate that our method is robust to the choice of D . For linear methods, we first regress the covariates X_1 and X_2 on each Y_i and then obtain the residuals $R_{k,i}$ for $k = 1, \dots, n$. Next, we compute the standardized residual covariance between each pair,

$$\hat{\xi}_{ij} = \frac{n\hat{\sigma}_{ij}}{\left[\sum_{k=1}^n \{(R_{k,i} - \bar{R}_i)(R_{k,j} - \bar{R}_j) - \hat{\sigma}_{ij}\}^2\right]^{1/2}},$$

where $\hat{\sigma}_{ij} = n^{-1} \sum_{k=1}^n (R_{k,i} - \bar{R}_i)(R_{k,j} - \bar{R}_j)$, $\bar{R}_i = n^{-1} \sum_{k=1}^n R_{k,i}$ and $\bar{R}_j = n^{-1} \sum_{k=1}^n R_{k,j}$. We then use the methods that were proposed in Cai and Liu (2016) to test whether Y_i and Y_j are linearly associated conditioning on covariates. Cai and Liu (2016) discussed two testing methods: one used the asymptotic null distribution of $\hat{\xi}_{ij}$ to perform testing (denoted by LIN-DEP), and the other used a bootstrap method to perform testing (denoted by LIN-DEP-B).

We ran 100 repetitions for each simulation setting. We considered two high dimensional scenarios $(n, p) = (300, 100)$ and $(n, p) = (300, 1000)$. Under all settings, we included 30 dependent pairs among a total of $p(p - 1)/2$ pairs. The results are summarized by the average empirical FDRs $\widehat{\text{FDR}}$ and false non-discoveries $\widehat{\text{FN}}$ across 100 repetitions. The optimal FDR is 0.05. Because there are 30 dependent pairs of variables in all settings, the range of $\widehat{\text{FN}}$ is 0–30, with 0 being the optimal value.

Table 2 displays the results for all five methods. Under setting 1, LIN-DEP and LIN-DEP-B outperform the SQUAC method by having higher power. Under settings 2–4 and 6, the SQUAC

Table 2. Simulation results for testing conditional dependence†

Method	Results for the following settings:											
	Setting 1		Setting 2		Setting 3		Setting 4		Setting 5		Setting 6	
	$\widehat{\text{FDR}}$	$\widehat{\text{FN}}$	$\widehat{\text{FDR}}$	$\widehat{\text{FN}}$	$\widehat{\text{FDR}}$	$\widehat{\text{FN}}$	$\widehat{\text{FDR}}$	$\widehat{\text{FN}}$	$\widehat{\text{FDR}}$	$\widehat{\text{FN}}$	$\widehat{\text{FDR}}$	$\widehat{\text{FN}}$
<i>n</i> = 300, <i>p</i> = 100												
SQUAC(3)	0.05	6.41	0.04	9.99	0.04	0.04	0.05	9.56	0.06	0.83	0.05	0.05
SQUAC(4)	0.06	7.08	0.05	11.11	0.06	0.00	0.05	6.70	0.06	0.00	0.05	0.00
SQUAC(5)	0.07	8.07	0.06	12.66	0.06	0.00	0.06	5.87	0.06	0.00	0.05	0.00
LIN-DEP-B	0.05	3.53	0.16	28.26	0.97	29.97	0.99	29.99	0.05	20.09	1.00	30.00
LIN-DEP	0.05	3.66	0.15	28.59	0.97	29.97	0.99	29.99	0.05	20.09	1.00	30.00
<i>n</i> = 300, <i>p</i> = 1000												
SQUAC(3)	0.05	10.91	0.05	14.90	0.05	0.42	0.04	15.14	0.05	6.47	0.04	0.45
SQUAC(4)	0.05	11.22	0.06	15.95	0.05	0.00	0.06	10.14	0.05	0.14	0.06	0.00
SQUAC(5)	0.05	12.49	0.06	16.92	0.07	0.00	0.05	9.20	0.06	0.00	0.06	0.00
LIN-DEP-B	0.04	7.71	0.49	29.49	1.00	30.00	1.00	30.00	0.05	20.67	1.00	30.00
LIN-DEP	0.03	7.93	0.51	29.33	1.00	30.00	1.00	30.00	0.03	20.76	1.00	30.00

† $\widehat{\text{FDR}}$, the average empirical FDR across 100 repetitions; $\widehat{\text{FN}}$, the average number of false non-discoveries across 100 repetitions.

Table 3. Simulation results for testing marginal dependence†

Results for the following settings:											
Setting 1		Setting 2		Setting 3		Setting 4		Setting 5		Setting 6	
$\widehat{FDR}$	$\widehat{FN}$	$\widehat{FDR}$	$\widehat{FN}$	$\widehat{FDR}$	$\widehat{FN}$	$\widehat{FDR}$	$\widehat{FN}$	$\widehat{FDR}$	$\widehat{FN}$	$\widehat{FDR}$	$\widehat{FN}$
<i>n</i> = 300, <i>p</i> = 100											
SQUAC(3)	0.05	6.49	0.04	10.19	0.04	0.04	0.05	8.78	0.05	0.78	0.04
LIN-DEP-B	0.05	1.72	0.30	28.66	0.98	29.98	0.99	29.99	0.05	20.00	1.00
LIN-DEP	0.05	1.76	0.28	29.07	0.98	29.98	0.99	29.99	0.05	20.00	1.00
Kendall	0.05	1.84	0.05	5.35	0.95	29.95	0.87	29.87	0.05	19.91	0.95
Spearman	0.05	1.84	0.05	5.35	0.96	29.96	0.91	29.91	0.06	19.96	0.96
<i>n</i> = 300, <i>p</i> = 1000											
SQUAC(3)	0.05	10.92	0.05	15.42	0.04	0.30	0.04	12.74	0.04	6.49	0.04
LIN-DEP-B	0.05	4.70	0.84	29.82	1.00	30.00	1.00	30.00	0.05	20.03	1.00
LIN-DEP	0.04	4.84	0.84	29.72	1.00	30.00	1.00	30.00	0.04	20.03	1.00
Kendall	0.04	4.73	0.05	10.26	1.00	30.00	0.97	29.97	0.04	20.00	1.00
Spearman	0.05	4.69	0.06	10.19	1.00	30.00	0.99	29.99	0.06	20.00	1.00

† $\widehat{FDR}$, the average empirical FDR across 100 repetitions; $\widehat{FN}$, the average number of false non-discoveries across 100 repetitions.

method performs satisfactorily whereas LIN-DEP and LIN-DEP-B can no longer control the FDR and suffer from low power. Under setting 5, although the linear methods LIN-DEP and LIN-DEP-B can control $\widehat{FDR}$ under the level desired, their power is much worse than those of the SQUAC methods. These results confirm the powerful performance of SQUAC methods in the presence of covariates, outliers and complicated dependence.

Table 2 also shows that the SQUAC method has robust performance with respect to *D*. Both their $\widehat{FDR}$ and their $\widehat{FN}$ are similar across *D* = 3, 4, 5. For settings 3–6, the false non-discoveries $\widehat{FN}$ for *D* = 4 or *D* = 5 are smaller than those for *D* = 3. However, under most settings, the FDR levels $\widehat{FDR}$ for *D* = 4 or *D* = 5 are larger than those for *D* = 3 and sometimes slightly inflated (e.g. empirical $\widehat{FDR}$ = 0.06 whereas the desired level is 0.05). As a common practice in hypothesis testing, the validity of multiple testing is more important than its power. We therefore recommend the use of *D* = 3 with a small-to-moderate sample size.

The SQUAC method is very efficient in computation. For *p* = 1000 and *n* = 300, one repetition of the SQUAC method takes about only 1 min on a Quad core machine with Intel Core i7-4771 3.5 Hz central processor unit and 16 Gbytes memory.

4.2. Numerical experiments on testing marginal quantile associations

Next, we conduct numerical experiments to demonstrate that the SQUAC method outperforms other competitive methods because of its ability to measure complex associations. We do not consider covariates here and compare the SQUAC method with LIN-DEP, LIN-DEP-B and two other multiple-testing methods based on Kendall’s τ -coefficient (Kendall) and Spearman’s ρ -coefficient (Spearman). Both these methods have been newly developed by us to test the rank associations. The details of these two multiple-testing methods are presented in the on-line supplementary materials.

Table 3 displays the comparison results. Under setting 1, where the dependence between variables is linear, LIN-DEP and LIN-DEP-B perform the best. The SQUAC method has worse

$\widehat{FN}$ than those of other methods but controls the FDR well. Setting 2 evaluates whether the methods are robust to a moderate percentage of outliers. LIN-DEP and LIN-DEP-B perform unsatisfactorily in this case, and all other methods still work reasonably well. Settings 3–5 examine the performance of these methods when complex association is present. Under both setting 3 and setting 4, the SQUAC method is the only method that can still control the FDR, while maintaining relatively low $\widehat{FN}$. The four other methods miss almost all dependent pairs and detected many false signals. Under setting 5, although all methods can control the FDR, they are not as powerful as the SQUAC method. Setting 6 simulates heavy tail random variables, where the SQUAC method significantly outperforms other methods in both FDR control and power. The results with $p = 1000$ are slightly worse than those with $p = 100$ for all five methods, but the relative performance patterns of these five methods remain the same. From these results, we observe that the SQUAC method is quite robust and can achieve adequate power under various settings, including in the presence of realistic complications such as outliers, non-linear associations and heavy-tailed random variables.

We did not compare our methods with other estimates that measure complex associations because their asymptotic null distributions are usually difficult to derive. Consequently, it is difficult to construct a multiple-testing method to control the FDR.

5. Data analysis of a gastric cancer study

We illustrate the method proposed by using the gastric cancer gene expression data that were generated by Lee *et al.* (2014). The full data set can be downloaded from <https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE26253>. It contains 432 patients diagnosed with gastric cancer: 68 of them are in stage I, 167 are in stage II, 130 are in stage III and 67 are in stage IV. Their 5-year disease-free survival time and recurrence status were also collected. Previous studies have shown that, although gastric cancer survival benefits from adjuvant chemotherapy or chemoradiation therapy (Noh *et al.*, 2014; Bang *et al.*, 2012), the 5-year disease-free survival rate remains poor for the patients who are diagnosed with late stage (III or IV) gastric cancer. In contrast, the patients who are diagnosed with early stage (I or II) gastric cancer have a better 5-year disease-free survival rate (Kim *et al.*, 2005; Lee *et al.*, 2004).

In this analysis, we separate the data set into two: one with early stage cancer patients ($n = 235$), and the other with late stage cancer patients ($n = 197$). We focus on the 85 genes belonging to the transforming growth factor (TGF) β signalling pathway because it plays a complex role in carcinogenesis, having both tumour suppressors and promoters (de Caestecker *et al.*, 2000). The gene identifiers and pathway information are available from the *Kyoto Encyclopedia of Genes and Genomes* database: http://www.genome.jp/kegg-bin/show_pathway?hsa04350. Our goal is to investigate the gene coexpression network for the early stage and late stage patients and further to infer the differential network.

In each data set (early or late stage), we fitted the quantile regression model (1) with the exact stage (I, II, III or IV) as the covariate and applied the proposed multiple-testing procedure (6) to identify conditional association between the pairs of gene expressions. The quantile points are set at $\tau_1 = 0.33$ and $\tau_2 = 0.67$, and the FDR is set at 5%. For each data set, we infer a gene coexpression network of p genes. If the conditional independence is rejected, we draw an edge between the two corresponding genes. Our method identifies 1313 edges in the early stage graph and 1098 edges in the late stage graph. We claim a differential edge if this edge exists in one graph but does not exist in the other. The multiple-testing procedure proposed identified 827 differential edges.

Table 4. Number of differential edges

Method	SQUAC	Kendall	Spearman	LIN-DEP	LIN-DEP-B
Number	827	727	733	731	66

As a comparison, we applied the alternative methods to the same data sets. For linear methods (the regression version of LIN-DEP and LIN-DEP-B), we used the exact stage as the covariate. For rank-association-based methods (Kendall and Spearman), we do not specify any covariate because they cannot adjust for any. The FDR is set at 5%. The numbers of differential edges that are identified by each method are displayed in Table 4. Our method detected more differential edges than did the alternative methods.

One way to check the validity of the differential edges detected is to examine the association between each differential edge and the survival outcome. Specifically, if the edge between Y_i and Y_j is differential, it is likely that the joint distribution of (Y_i, Y_j) will affect the survival outcome. To check this point, we combine the early stage patients and late stage patients into one data set. For subject k , $k = 1, 2, \dots, 432$, we use an eight-dimensional indicator vector to summarize the location of his or her paired outcome (Y_{ki}, Y_{kj}) in the joint distribution, accounting for disease stage. Specifically, let $\mathbf{S}_k = (S_{k1}, S_{k2}, S_{k3})'$ be the dummy variable of the four stages of the k th patient. Let $\hat{q}_{is}(\mathbf{S}_k)$ ($s = 0, 1, 2, 3; i = 1, \dots, p$) be the estimated marginal τ_s -quantile of Y_i , where $\hat{q}_{i0}(\mathbf{S}_k) = -\infty$, and $\hat{q}_{i3}(\mathbf{S}_k) = \infty$. We set $J_{k,ij,t}$ as an indicator of the conditional quantile cell that (Y_{ki}, Y_{kj}) falls into:

$$J_{k,ij,d} = I\{Y_{ki} \in (\hat{q}_{i,s_1-1}(\mathbf{S}_k), \hat{q}_{i,s_1}(\mathbf{S}_k)], Y_{kj} \in (\hat{q}_{j,s_2-1}(\mathbf{S}_k), \hat{q}_{j,s_2}(\mathbf{S}_k)]\}, \quad d = 3(s_1 - 1) + s_2.$$

It is easy to see that $\sum_{d=1}^9 J_{k,ij,d} = 1$. Let $\mathbf{J}_{k,ij} = (J_{k,ij,2}, \dots, J_{k,ij,9})'$ be the eight-dimensional covariate. If the differential edge between gene i and gene j is associated with survival, at least one $J_{k,ij,d}$ ($d = 2, \dots, 9$) should be associated with the survival rate. Then, we fit a Cox proportional hazard model $\lambda(v|\mathbf{J}_{k,ij}) = \lambda_{0,ij}(v) \exp(\sum_{d=1}^8 \beta_{ij,d} J_{k,ij,d+1})$, where $\lambda(v)$ is the hazard function, and $\lambda_{0,ij}(v)$ is the baseline hazard function that both gene i and gene j fall into the lowest sample tertile. We set the null hypothesis $H_{0,ij} : \beta_{ij,d} = 0, \forall d = 1, \dots, 8$, and derived its corresponding p -values and their estimated density. For each method, we first determine the set of differential edges and then perform the above procedure to obtain the estimated density curve (presented in Fig. 1). It is clear that the p -values corresponding to the SQUAC method tend to take smaller values, indicating that the paired outcomes (e.g. edges) detected by the SQUAC method are more likely to be associated with the survival outcomes.

To illustrate the application of the proposed method further, we count the degree (the number of edges stemming from one vertex) of each gene in early and late stage gene coexpression networks identified by the SQUAC method and calculate the degree differences between two networks. The top five genes with the largest degree differences are *TGF- β 3*, *BMP4*, *DCN*, *AMHR2* and *SMAD4*. *TGF- β 3*, as a TGF β family member, has been discovered to play a dual role in human cancer in different prognostic stages of cancer—in early stages, it performs as a tumour suppressor, whereas in the late stages it performs as a tumour promotor (Jakowlew, 2006; Lebrun, 2012). *BMP4* has been identified as a modulator of cisplatin (a widely used gastric cancer chemotherapy) sensitivity in gastric cancer (Ivanova *et al.*, 2013). *DCN* is capable of suppressing the growth of multiple types of tumour. For example, evidence has shown that it is a key regulator for chemoresistant mechanisms for oral cancer (Kasamatsu *et al.*, 2015) and

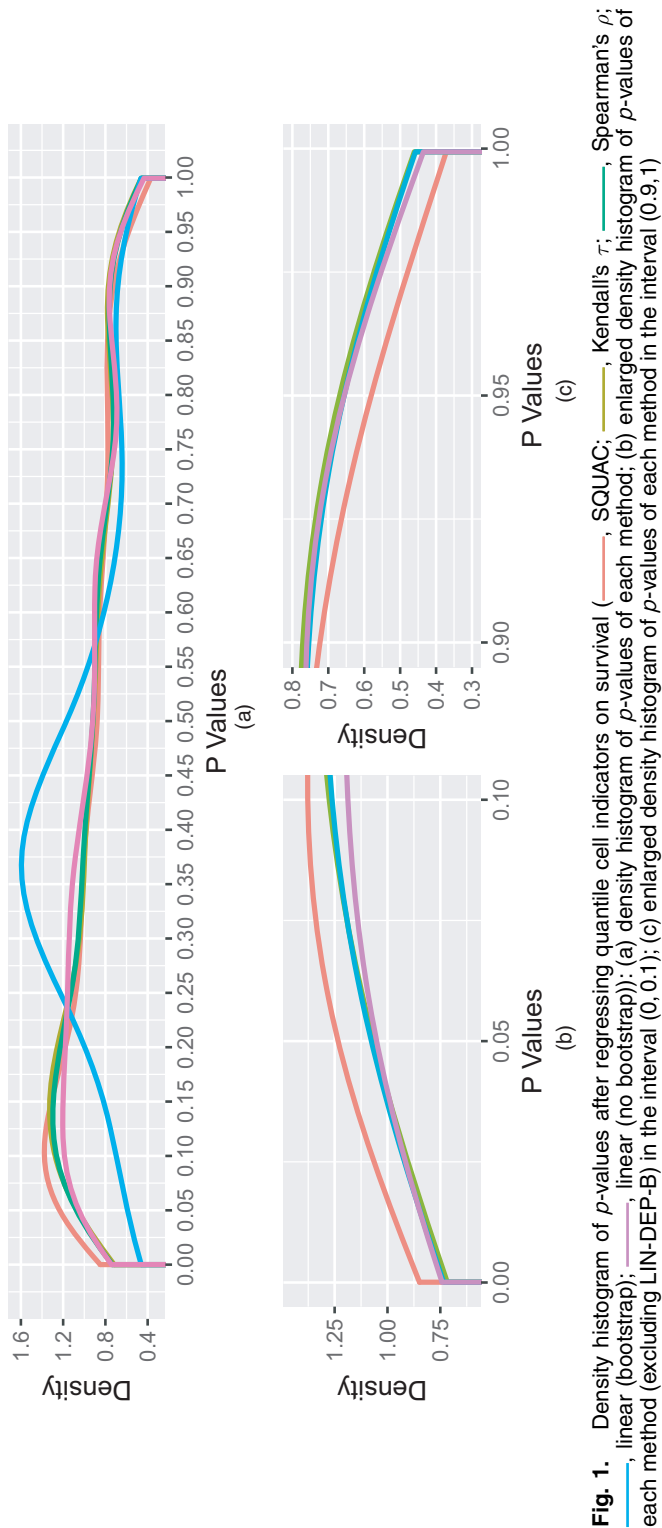


Fig. 1. Density histogram of p -values after regressing quantile cell indicators on survival (—): linear (no bootstrap); (—): linear (bootstrap); (—): LIN-DEP-B; (—): LIN-DEP-A; (—): SQUAC; (—): Kendall's τ ; (—): Spearman's ρ ; each method (excluding LIN-DEP-B) in the interval (0, 0.1); (c) enlarged density histogram of p -values of each method (excluding LIN-DEP-B) in the interval (0.9, 1)

is associated with breast cancer metastasis and survival (Cawthorn *et al.*, 2012; Ishiba *et al.*, 2014). *AMHR2* has been recently shown to regulate survival signalling in non-small cell lung cancer (Beck *et al.*, 2015). *SMAD4* has been shown to be significantly related to the prognostic differences in gastric cancer patients as well as to the survival rates (Leng *et al.*, 2009; Wang *et al.*, 2007). This result demonstrates the practical utility of the SQUAC method in real life biomedical studies.

6. Extension to quantile regression in high dimensional sparse models

In genetics and genomics study, high dimensional covariates may affect the levels of expression. For example, in expression quantitative trait loci studies, in addition to gene expression levels, thousands or even millions of genetic markers are measured. Researchers are interested in the gene coexpression patterns conditioning on the genetic markers. The null hypotheses are $H_{0,ij}: Y_i \perp\!\!\!\perp Y_j | \mathbf{X}$, where Y_i and Y_j are expression levels of gene i and gene j , and $\mathbf{X} = (X_1, \dots, X_{p_x})'$ are genetic markers. We use the same quantile regression model (1). Since only a few genetic markers will affect the gene expression level of a certain gene, we assume that $\beta_{0,i}(\tau_s)$ is a sparse vector for all genes. Let $s_x = \sup_{1 \leq i \leq p, 1 \leq s \leq D-1} \|\beta_{0,i}(\tau_s)\|_0$. It represents the maximum level of sparsity of all quantile coefficient vectors, which is usually much smaller than n .

When sparse high dimensional covariates exist, the traditional coefficient estimation method (2) no longer works; penalized regression methods are usually used to yield consistent estimators. Many references have discussed such procedures. See, for example, Belloni and Chernozhuko (2011), Wang *et al.* (2012), Zheng *et al.* (2013, 2015) and Fan *et al.* (2014). These references also propose different conditions and the rate of convergence of quantile levels that can be achieved under these conditions.

Considering these rates of convergence under the case $p_x \rightarrow \infty$, we propose to adjust conditions 2 to make sure that the main results hold for the high dimensional sparse covariates setting.

Condition 2'. Assume that $p_x \leq cn^r$ for some $r > 0$, s_x is a constant and there is a constant C_2 such that

$$P\left\{\sup_{s=1}^D |\hat{Q}_{k,i,s} - Q_{k,i,s}| > 2C_2(1+\varepsilon)n^{-1/2} \log(n \vee p)^{1/2}\right\} < D(n \vee p)^{-2-\varepsilon}. \quad (15)$$

Some existing methods enjoy the property that is described in condition (15) under certain conditions, such as the adaptive robust lasso method that was developed in Fan *et al.* (2014). A description of why this method yields estimates satisfying condition 2' is provided in section S5 of the on-line supplementary materials.

7. Discussion

In this paper, we use quantile associations to measure the general association between variables and to develop the SQUAC test statistics to measure the quantile dependence. The proposed testing and multiple-testing procedures are based on the SQUAC statistics. Commonly used surrogates for associations, such as linear associations (correlation) and rank associations (Kendall's τ - or Spearman's ρ -coefficients), require parametric or semiparametric assumptions to hold to capture fully the dependence between variables. Quantile associations require considerably weaker assumptions. They require neither assumptions on the parametric form of

the distributions nor on the association structures. Therefore, they are robust and sufficiently flexible to be applied to measure complicated associations.

When deriving quantile associations, the quantile levels and their number D are predetermined. In genomic applications, it is reasonable to use tertile points to cut gene expression levels because it is easy to interpret the results. On the basis of our simulation results, we recommend using $D = 3$ for small-to-moderate sample sizes. In general, as D increases, the SQUAC statistic can capture more local associations and become increasingly more powerful to test general associations. In contrast, large D will slow down the convergence of SQUAC statistics' asymptotic null distribution. For simplicity, we use only even grids of quantile levels for our method. Nevertheless, both the value of D and the quantile points will affect the amount of quantile association that is captured by the method proposed, and therefore will affect its power. For different pairs of (Y_i, Y_j) , the value of D and the quantile levels can be set differently and adaptively to capture weak general associations. Because the distribution of $(Y_1, \dots, Y_p)'$ is usually unknown, the derivation of the optimal D and the quantile levels is not easy. We shall investigate the data-adaptive method to choose their optimal values in the future.

The current theoretical results of the SQUAC method are based on the linear quantile regression model assumption to obtain appropriate conditional quantile estimates. It is possible to extend the idea to models beyond linear quantile regression models towards more general models, such as non-parametric quantile regression models or even complete non-parametric models. Theoretically, without any model assumption of each Y_i and $\mathbf{X}$, one can still estimate the conditional distribution function in a non-parametric framework to obtain the estimate of the quantiles. As long as conditions 1 and 2 are satisfied, the theoretical results can be extended for more general cases.

8. Supplementary materials

We include in the on-line supplementary materials the connection and difference between our method and other quantile association inference methods, the proofs of the lemmas, a detailed description of the six simulation settings, the multiple-testing methods Kendall and Spearman, and a description on why the adaptive lasso method that was developed in Fan *et al.* (2014) yields estimates that satisfy condition 2'.

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Appendix A: Proofs of the theorems

For all k, i and s , define

$$\hat{I}_{k,i,s} = I(\hat{Q}_{k,i,s-1} < Y_{ki} \leq \hat{Q}_{k,i,s}),$$

$$I_{k,i,s} = I(Q_{k,i,s-1} < Y_{ki} \leq Q_{k,i,s}).$$

An equivalent form of the SQUAC statistic is

$$T_{ij} = \sum_{s=1}^D \sum_{t=1}^D \left\{ n^{-1/2} \sum_{k=1}^n \frac{\hat{I}_{k,i,s} \hat{I}_{k,j,t} - \nu_s \nu_t}{(\nu_s \nu_t)^{1/2}} \right\}^2.$$

Lemmas 1–7 are required to prove the asymptotic properties of the method proposed. Among those, lemmas 1–5 are listed below, and lemmas 6 and 7 are listed in the proof of the theorems.

Lemma 1.

$$\Delta_{ij,st} = n^{-1/2} \sum_{k=1}^n \frac{\hat{I}_{k,i,s} \hat{I}_{k,j,t} - \nu_s \nu_t}{(\nu_s \nu_t)^{1/2}} - n^{-1/2} \sum_{k=1}^n \frac{(I_{k,i,s} - \nu_s)(I_{k,j,t} - \nu_t)}{(\nu_s \nu_t)^{1/2}}. \quad (16)$$

Then, for all $\varepsilon > 0$,

$$P \left[\sup_{(i,j) \in \mathbb{H}} \sup_{1 \leq s, t \leq D} |\Delta_{ij,st}| > (M_2 + \varepsilon) \log(n \vee p)^{1/2} \right] \leq CD(n \vee p)^{-\varepsilon/M}, \quad (17)$$

with $M_2 = (3 - 2u_0)(4C_1C_2 + 2)/u_0$ and some sufficiently large M .

Further, depending on positive integer m ,

$$\sup_{(i,j) \in \mathbb{H}_0} \sup_{1 \leq s, t \leq D} E(\Delta_{ij,st}) \leq Cn^{-1/2}, \quad (18)$$

$$\sup_{(i,j) \in \mathbb{H}_0} \sup_{1 \leq s, t \leq D} E(|\Delta_{ij,st}|^m) \leq C_m \{\log(n \vee p)^2/n\}^{m/2}, \quad (19)$$

where $C_m = \{4(1 + \varepsilon)C_1C_2\}^m m_0^{-m} m!$.

Lemma 2. Suppose that X follows a χ^2 -distribution with degree of freedom K , where K is a fixed positive integer. Then

$$\lim_{t \rightarrow \infty} \frac{P(X > t)}{\Gamma(K/2)^{-1} (t/2)^{K/2-1} \exp(-t/2)} = 1. \quad (20)$$

Lemma 3. Consider $\mathbb{H}_{01}$ defined in expression (S28) in the on-line supplementary material. Under conditions 1–4, for any $1 \leq t \leq b_p$,

$$\sup_{(a,b,c,d) \in \mathbb{H}_{01}} |P(T_{ab} > t, T_{cd} > t) - P(T_{ab} > t)P(T_{cd} > t)| = o\{G_D(t)^2\}. \quad (21)$$

Lemma 4. Let

$$\tilde{T}_{ij} = \sum_{s=1}^D \sum_{t=1}^D \left\{ n^{-1/2} \sum_{k=1}^n \frac{(I_{k,i,s} - \nu_s)(I_{k,j,t} - \nu_t)}{(\nu_s \nu_t)^{1/2}} \right\}^2.$$

Then

$$\sup_{(i,j) \in \mathbb{H}_0} \left| \frac{P(\tilde{T}_{ij} \geq t)}{G_D(t)} - 1 \right| \leq CD^6(1+t)^{3/2}n^{-1/2}, \quad \forall t \text{ satisfying } t = o(n^{1/3}D^{-4}). \quad (22)$$

Lemma 5. For any positive integer D , let ϕ_{I_D} be the density of the D -dimensional standard multivariate Gaussian distribution, and $Q_K(u)$ be a K th order polynomial of $u \in \mathbb{R}^D$ with bounded coefficients. Then, for any $t \geq 0$, $|\int_{\|u\|^2 \geq t} Q_K(u) \phi_{I_D}(u) du| \leq C_K D^K (1+t)^{K/2} \int_{\mathbb{R}^D} f_{\chi_D^2}(x) dx$, where C_K is a constant depending on K only.

A.1. Proof of theorem 1

Let

$$L_{ij,st} = n^{-1/2} \sum_{k=1}^n \frac{(I_{k,i,s} - \nu_s)(I_{k,j,t} - \nu_t)}{(\nu_s \nu_t)^{1/2}}. \quad (23)$$

Recall $\Delta_{ij,st}$ defined in equation (16). Then $T_{ij} = \tilde{T}_{ij} + R_{ij}$, where

$$\begin{aligned}\tilde{T}_{ij} &= \sum_{s=1}^D \sum_{t=1}^D L_{ij,st}^2, \\ R_{ij} &= 2 \sum_{s=1}^D \sum_{t=1}^D L_{ij,st} \Delta_{ij,st} + \sum_{s=1}^D \sum_{t=1}^D \Delta_{ij,st}^2.\end{aligned}\quad (24)$$

When $j_1 \neq j_2$, $E(I_{k,i,j_1} I_{k,i,j_2}) = 0$, and, when $j_1 = j_2$, $E(I_{k,i,j_1} I_{k,i,j_2}) = E(I_{k,i,j_1}) = \nu_{j_1}$. Then, for any two pairs (s_1, t_1) and (s_2, t_2) , it is easy to see that

$$\text{cov}(L_{ij,s_1,t_1}, L_{ij,s_2,t_2}) = \begin{cases} (1 - \nu_s)(1 - \nu_t), & \text{if } s_1 = s_2 = s, t_1 = t_2 = t, \\ -(1 - \nu_s)(\nu_{t_1} \nu_{t_2})^{1/2}, & \text{if } s_1 = s_2 = s, t_1 \neq t_2, \\ -(\nu_{s_1} \nu_{s_2})^{1/2}(1 - \nu_t), & \text{if } s_1 \neq s_2, t_1 = t_2 = t, \\ (\nu_{s_1} \nu_{s_2})^{1/2}(\nu_{t_1} \nu_{t_2})^{1/2}, & \text{if } s_1 \neq s_2, t_1 \neq t_2. \end{cases} \quad (25)$$

Define random vector $\mathbf{L}_{ij} = (L_{ij,11}, L_{ij,12}, \dots, L_{ij,DD})'$. By expression (25), $\text{var}(\mathbf{L}_{ij}) = \Sigma = \Sigma_1 \otimes \Sigma_1$, where $\Sigma_1 = \mathbf{I}_D - \sqrt{\nu} \sqrt{\nu}'$, with $\sqrt{\nu} = (\sqrt{\nu_1}, \dots, \sqrt{\nu_D})'$. By the central limit theorem, $\mathbf{L}$ asymptotically follows the multivariate normal distribution $N_{D^2}(\mathbf{0}, \Sigma_1 \otimes \Sigma_1)$.

Let $\Sigma_1 = (\sigma_{ij})_{D \times D}$ and $\Sigma = \Sigma_1 \otimes \Sigma_1$. Because $\sqrt{\nu}' \sqrt{\nu} = \sum_{i=1}^D \nu_i = 1$, Σ_1 is idempotent, i.e. $\sum_{k=1}^D \sigma_{ik} \sigma_{kj} = \sigma_{ij}$. Then Σ is also idempotent, because

$$\Sigma^2 = (\sigma_{ij} \Sigma_1)_{D^2 \times D^2} \times (\sigma_{ij} \Sigma_1)_{D^2 \times D^2} = \left(\sum_{k=1}^D \sigma_{ik} \sigma_{kj} \Sigma_1^2 \right)_{D^2 \times D^2} = (\sigma_{ij} \Sigma_1)_{D^2 \times D^2} = \Sigma.$$

Because Σ_1 is idempotent and symmetric, Σ_1 is a projection matrix with $\text{rank}(\Sigma_1) = \text{tr}(\Sigma_1) = \sum_{i=1}^D (1 - \nu_i) = D - 1$. Therefore, $\text{rank}(\Sigma) = \text{rank}(\Sigma_1)^2 = (D - 1)^2$.

By theorem 5.5A in Rencher (2000), because Σ is idempotent with degrees of freedom $(D - 1)^2$, $\tilde{T}_{ij} = \mathbf{L}_{ij}' \mathbf{L}_{ij}$ asymptotically follows the central χ^2 -distribution with degrees of freedom $(D - 1)^2$.

Then

$$\begin{aligned}\sup_{0 \leq t \leq C_0 \log(n \vee p)} \sup_{(i,j) \in \mathbb{H}_0} \frac{P(T_{ij} > t)}{G_D(t)} &\leq \sup_{0 \leq t \leq C_0 \log(n \vee p)} \sup_{(i,j) \in \mathbb{H}_0} \frac{P\{\tilde{T}_{ij} \geq t - \log(n \vee p)^{-1}\} + P\{R_{ij} > \log(n \vee p)^{-1}\}}{G_D(t)} \\ &= \sup_{0 \leq t \leq C_0 \log(n \vee p)} \sup_{(i,j) \in \mathbb{H}_0} \frac{G_D\{t - \log(n \vee p)^{-1}\}}{G_D(t)} \frac{P\{\tilde{T}_{ij} > t - \log(n \vee p)^{-1}\}}{G_D\{t - \log(n \vee p)^{-1}\}} \\ &\quad + \frac{P\{R_{ij} > \log(n \vee p)^{-1}\}}{G_D(t)}\end{aligned}\quad (26)$$

by lemma 2 and lemma 4.

When t is a constant, based on the continuity of $G_D(t)$, for any small ε_p , when p sufficiently large,

$$|G_D\{t - \log(n \vee p)^{-1}\} / G_D(t) - 1| \leq \varepsilon_p. \quad (27)$$

When $t = O\{\log(n \vee p)\}$, lemma 2 leads to inequality (27).

Lemma 4 leads to

$$\left| \frac{P\{\tilde{T}_{ij} > t - \log(n \vee p)^{-1}\}}{G_D\{t - \log(n \vee p)^{-1}\}} - 1 \right| \leq CD^6 \log(n \vee p)^{3/2} n^{-1/2} \leq \varepsilon_p.$$

To prove the final result, we also need the following lemma.

Lemma 6. When $0 \leq t \leq C_0 \log(n \vee p)$, $P\{|R_{ij}| > \log(n \vee p)^{-1}\} / G_D(t) \leq \varepsilon_p$.

Thus,

$$\sup_{0 \leq t \leq C_0 \log(n \vee p)} \sup_{(i,j) \in \mathbb{H}_0} \frac{P(T_{ij} > t)}{G_D(t)} \leq 1 + 3\varepsilon_p.$$

Similarly, by

$$\sup_{0 \leq t \leq C_0 \log(n \vee p)} \sup_{(i,j) \in \mathbb{H}_0} \frac{P(T_{ij} > t)}{G_D(t)} \geq \sup_{(i,j) \in \mathbb{H}_0} \frac{P(\tilde{T}_{ij} \leq t + c) - P(R_{ij} > c)}{G_D(t)},$$

we can show that

$$\sup_{0 \leq t \leq C_0 \log(n \vee p)} \sup_{(i,j) \in \mathbb{H}_0} \frac{P(T_{ij} > t)}{G_D(t)} \geq 1 - 3\varepsilon_p.$$

This completes the proof.

A.2. Proof of theorem 2

It is easy to see that

$$\sup_{1 \leq i, j \leq p} \sup_{1 \leq s, t \leq D} \left| \frac{(I_{k,i,s} - \nu_s)(I_{k,j,t} - \nu_t)}{(\nu_s \nu_t)^{1/2}} - \gamma_{ij,st} \right| \leq M_1,$$

by Azuma's inequality,

$$P \left\{ \sup_{1 \leq s, t \leq D} |L_{ij,st} - n^{1/2} \gamma_{ij,st}| > (2 + \varepsilon) M_1 \log(n \vee p)^{1/2} \right\} \leq 2D^2 (n \vee p)^{-(2+\varepsilon)}.$$

Because $\text{Card}(\mathbb{B}_{c_2}) \leq p^2$,

$$P \left\{ \sup_{(i,j) \in \mathbb{B}_{c_2}} \sup_{1 \leq s, t \leq D} |L_{ij,st} - n^{1/2} \gamma_{ij,st}| > (2 + \varepsilon) M_1 \log(n \vee p)^{1/2} \right\} \leq 2D^2 (n \vee p)^{-\varepsilon}.$$

$T_{ij} = \sum_{s=1}^D \sum_{t=1}^D (n^{1/2} \gamma_{ij,st} + L_{ij,st} - n^{1/2} \gamma_{ij,st} + \Delta_{ij,st})^2$. When $|\gamma_{ij,st}| \geq c_2 \{\log(n \vee p)/n\}^{1/2}$ and $c_1 = c_2 - 2M_1 - M_2 - \varepsilon > 0$,

$$\begin{aligned} P \left\{ \inf_{(i,j) \in \mathbb{B}_{c_2}} T_{ij} > c_1 \log(n \vee p) \right\} &\geq 1 - P \left\{ \sup_{(i,j) \in \mathbb{B}_{c_2}} \sup_{1 \leq s, t \leq D} |\Delta_{ij,st}| > (M_2 + \varepsilon/2) \log(n \vee p)^{1/2} \right\} \\ &\quad - P \left[\sup_{(i,j) \in \mathbb{B}_{c_2}} \sup_{1 \leq s, t \leq D} |L_{ij,st} - n^{1/2} \gamma_{ij,st}| \geq \{2 + \varepsilon/(2M_1)\} M_1 \log(n \vee p)^{1/2} \right] \\ &\geq 1 - CD^2 (n \vee p)^{-\varepsilon/M}. \end{aligned}$$

A.3. Proof of theorem 3

By the continuity of $G_D(t)$ and the monotonicity of $\sum_{i,j} I(T_{ij} > t)$, we can obtain that, for some $0 \leq \hat{t} \leq t_p$, $G(\hat{t})q / \max\{\sum_{1 \leq i < j \leq p} I(T_{ij} > \hat{t}), 1\} = \alpha$.

By theorem 2, $\sum_{i,j} I(T_{ij} > t_p) \geq c_p$. Let $b_p = t_p - 2 \log(c_p)$. By definition of $\hat{t}$, we have $P(0 \leq \hat{t} \leq b_p) \rightarrow 1$.

To prove theorem 3, it suffices to show that

$$\sup_{0 \leq t \leq b_p} \left| \frac{\sum_{(i,j) \in \mathbb{H}_0} I(T_{ij} > t)}{qG_D(t)} - 1 \right| \rightarrow 0 \quad \text{in probability.}$$

Let $e_p(t) = E[\sum_{(i,j) \in \mathbb{H}_0} I(T_{ij} > t) / \{qG_D(t)\}] = q_0 P(T_{ij} > t) / \{qG_D(t)\}$. By lemma 1, for $0 \leq t \leq b_p$, $e_p(t) \rightarrow 1$. By Chebyshev's inequality, $\forall \epsilon > 0$,

$$P \left\{ \left| \frac{\sum_{(i,j) \in \mathbb{H}_0} I(T_{ij} > t)}{qG_D(t)} - 1 \right| > \epsilon \right\} \leq \frac{\text{var} \left[\frac{\sum_{(i,j) \in \mathbb{H}_0} I(T_{ij} > t) / \{qG_D(t)\}}{\epsilon^2} \right] + \{e_p(t) - 1\}^2}{\epsilon^2}.$$

Therefore, it suffices to show that $\sup_{0 \leq t \leq b_p} \text{var}[\sum_{(i,j) \in \mathbb{H}_0} I(T_{ij} > t) / \{qG_D(t)\}] \rightarrow 0$.

$$\begin{aligned} \text{var} \left\{ \frac{\sum_{(i,j) \in \mathbb{H}_0} I(T_{ij} > t)}{qG_D(t)} \right\} &= \frac{1}{q^2 \{G_D(t)\}^2} \sum_{(i,j) \in \mathbb{H}_0} P(T_{ij} > t) P(T_{ij} \leq t) \\ &\quad + \frac{1}{q^2 G_D(t)^2} \sum_{(a,b), (c,d) \in \mathbb{H}_0, (a,b) \neq (c,d)} \{P(T_{ab} > t, T_{cd} > t) - P(T_{ab} > t) P(T_{cd} > t)\}. \end{aligned}$$

As p is sufficiently large, the first term

$$\frac{1}{q^2 G_D(t)^2} \sum_{(i,j) \in \mathbb{H}_0} P(T_{ij} > t) P(T_{ij} \leq t) \leq \frac{2}{\alpha c_p} \frac{P(T_{ij} > t)}{G_D(t)} \leq C c_p^{-1} \rightarrow 0.$$

Combined with lemma 7, we prove the result.

Lemma 7. As $n \vee p \rightarrow \infty$,

$$\frac{1}{q^2 G_D(t)^2} \sum_{(a,b,c,d) \in \mathbb{H}_{02}} \{P(T_{ab} > t, T_{cd} > t) - P(T_{ab} > t) P(T_{cd} > t)\} \rightarrow 0.$$

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Supporting information

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